Diabetic Retinopathy Detection: An Automatic Modelling Using Artificial Intelligence & Deep Learning

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***Abstract*— Diabetic retinopathy (DR) is a leading cause of vision loss among individuals with diabetes. Early detection and timely treatment of DR are crucial to prevent irreversible damage. In recent years, artificial intelligence (AI) and deep learning techniques have shown promising results in automating the detection of DR from retinal images. This paper presents an automatic modeling approach for diabetic retinopathy detection, leveraging the power of AI and deep learning. The proposed method utilizes a convolutional neural network (CNN) architecture designed to analyze retinal images and classify them into different stages of diabetic retinopathy. To evaluate the performance of the proposed model, extensive experiments are conducted on a publicly available diabetic retinopathy dataset. The model's accuracy, sensitivity, specificity, and other performance metrics are assessed using standard evaluation protocols. The results demonstrate that the proposed automatic modeling approach achieves highly accurate and reliable diabetic retinopathy detection. The model shows excellent performance in distinguishing different stages of DR, enabling early intervention and appropriate treatment strategies. The implications of this research are significant, as the automatic detection of diabetic retinopathy using AI and deep learning techniques can aid healthcare professionals in the early diagnosis and management of the disease.**

Keywords Diabetic Retinopathy, Artificial Intelligence, Deep Learning, Retinal Images

I. Introduction

The International Diabetes Federation (IDF) gave data that more than 530 million adults are living with diabetes, which is projected to rise by 643 million by 2023. It says diabetes is the major cause of more than 6.7 million deaths in 2021 [1]. High levels of blood sugar give rise multiple conditions that are typically grouped together and, condition is termed as Diabetes. Majorly, the condition is categorized as type 1, which arises when the human body does not make enough insulin; type 2, which arises when insulin cannot be properly utilized. In both of the conditions sugar levels in the human body starts to rise, leading to the sugar accumulation. The first condition that arises is pre-diabetic when levels rise the threshold value, and this condition is not termed as diabetes [3]. Low- and middle-income nations account for almost three out of every four persons with diabetes. It has become a big concern in healthcare since the rising prevalence of diabetes mostly due to metabolomic variation, not only impairs the quality of life of those who have the condition, but also places a significant fiscal strain on our healthcare system. Consequences of diabetes, which include blindness, amputation, kidney failure, heart attack, stroke, and an increased risk of certain forms of cancer, such as liver, pancreatic, uterine, colon, breast, and bladder cancer, account for a significant portion of its economic burden [1,4].

[1] Diabetes mellitus causes vascular complications leading to irreversible loss of vision, this condition is termed as Diabetic retinopathy (DR), diagnosed in almost one-third of the patients with high levels of sugar. DR can be categorised as non-proliferative retinopathy (NPDR) and proliferative retinopathy (PDR). Blood vessels leak in the retina during NPDR, considered as an early disease stage. This presents as fluid, hemorrhage, or lipid in the retina causing blood veins shut, finally resulting to condition such as ischemia or inadequate blood flow. This condition on increasing the severity causes PDR, condition in which malfunctioning blood vessels are increased, due to ischemia. Blood starts leaking from these veins and blood reaches the vitreous (a gel-like substance) and may cause the retina surface to behave abnormally experiencing the tractional changes that eventually cause the retina to separate and complete vision loss [5]. Risk factors the causing DR are Blood Pressure, High Cholesterol, Blood Sugar Level, Gender, Pregnancy, Tobacco, Smoking, Age, BMI, And Hba1c [6].

The retinal vascular illness gives rise to increased vascular permeability and diffuse retinal ischemia. In United states DR and DR leading to vision loss among the diabetic population is forecasted as 28.5 % and 4.4 % respectively. Early detection, frequent eye exams, aggressive control of

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hyperglycemia and hypertension, and medicinal and/or surgical ocular therapy are all part of the DR management process. It is evident since last two decades that early diagnosis and proper treatment can revert the vision loss upto 50% in DR patients [7,8].

II. DR Diagnosis

Clinically DR severity level is classified into five grades, as discussed in Table 1.

Table I: Classification of DR Diagnosis

Proposed Disease Severity Level	**Findings Observable Upon Retina Examination**
No apparent retinopathy	No abnormalities
Mild Non-Proliferative Diabetic Retinopathy (Mild NPDR)	Microaneurysms only
Moderate Non-Proliferative Diabetic Retinopathy (Mod NPDR)	More than just microaneurysms but less than Severe NPDR
Severe Non-Proliferative Diabetic Retinopathy (NPDR)	• Identifiable venous beading in 2+ quadrants; • More than 20 intraretinal hemorrhages in each of the 4 quadrants; • Prominent IRMA in 1+ quadrant
Proliferative Diabetic Retinopathy (PDR)	• Hemorrhage in the vitreous/preretinal area • Neovascularization

In the case of DR, early symptoms are very rare, though they are typically diagnosed in the late stages. Some of the most common symptoms include blank or dark areas in vision spot, blurred vision, spots, dots, or dark strings floating, poor night vision, and vision loss. Retinal (fundus) photography, a manually interpreted format, has been widely acknowledged as a screening tool for diabetic retinopathy, with exceptional performance that can outperform in-person dilated eye exams. Thus, it creates a plethora of opportunities for creating an automatic tool for grading the severity of diabetic retinopathy as it can enhance and accelerate the detection and treatment of patients who might be containing these diseases, unaware of their presence within them. Artificial Intelligence, Machine learning, and Deep learning are the systems that can help to achieve this goal most effectively. the current study focuses on using machine learning and deep learning tools and algorithms to develop an automated system for DR screening.

III. AI-ML and Deep Learning as An Approach Toward DR Detection

Artificial intelligence (AI) is one of the most powerful and trending systems which mimics human intelligence to perform human tasks, which can improve itself over time and gradually as it collects more information or data in layman terms. In AI there are two subfields: machine learning (ML) and deep learning (DL). They include highly computational methods having potential in identifying diabetic retinopathy with accuracy.

These powerful and complex algorithms make it easy to understand and interpret that data for accurate predictions irrespective of the type of data image, tabular or excel, etc. Using such methodologies for biological development opens the scope of finding various solutions to a biological problem which are logically justified and could be used for verification on clinical scenarios.

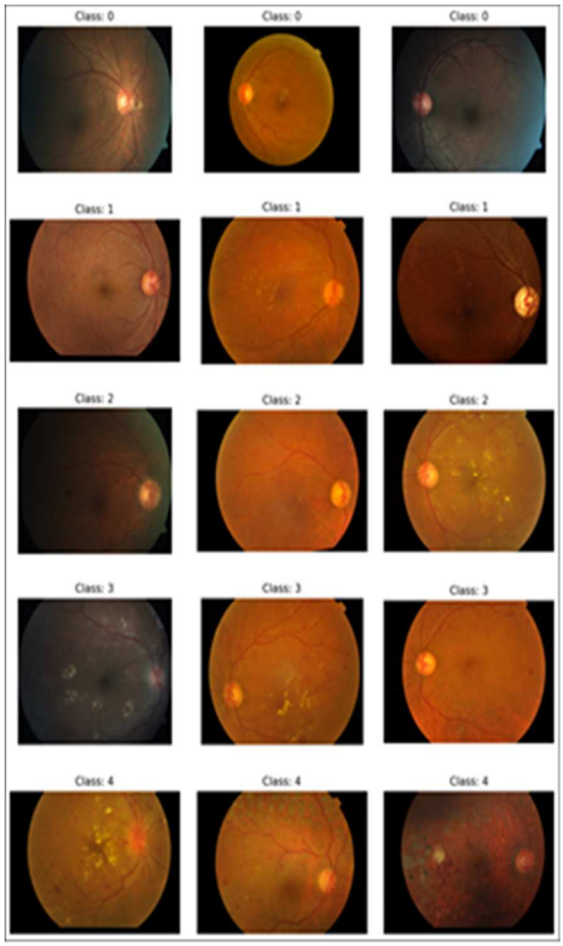

Fig 1. Reshaped and normalized, image samples are labeled with information on the existence of eye illness, ranging from 0 to 4 on a severity scale.

A collection of images that has been tagged with a subject ID, as well as left or right, and the presence of diabetic retinopathy in each image on a scale of 0 to 4, where 0, No DR Diabetic retinopathy is not evident in the retina. Lesions or anomalies are not visible. 1 mild DR, there are early

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indications of diabetic retinopathy. This could involve tiny patches of enlargement in the blood vessels of the retina (microaneurysms), 2 moderate degrees of DR More obvious alterations in the retina include bleeding, an increase in microaneurysms, and potentially some blood vessel blockage. 3 severe DR, Retinal injury of considerable magnitude with several blood vessels occluded. This may result in 4 proliferative DR, the formation of new, aberrant blood vessels, and inadequate blood flow to certain regions of the retina. the most developed stage of retinopathy caused by diabetes. On the surface of the retina, new blood vessels begin to form. This can cause retinal detachment and blindness, among other severe visual issues. These publicly available datasets were used in the analysis from Kaggle datasets [9]. The data size is 3662 images, which is further resized and normalized. Figure 1 Represents the image samples after processing.

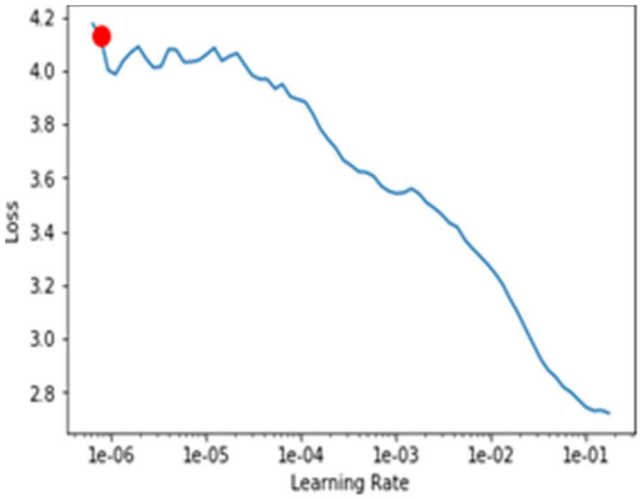

Fig 2. Loss vs. Learning Rate for sample data shows that the loss decreases faster around 1r = 1e-2; hence, this can be used for training the model.

Further, in the way of understanding the framework of our detection model, normalizing images as features are must, although, in general, there is a tradeoff between simplicity and the incorporation of affluent information in clinical decision-making. Better features and strategies prove to be more valuable for patients as added complexity of additional information enlarges the likelihood of good decision-making and improved outcomes, and handling the complexity without increasing the calculation burden for clinicians becomes imperative, which justifies the use of deep learning algorithms at best because of algorithms and tools which contain such high computable efficiency.

Here we have utilized transfer learning for the implementation of our model for the detection of disease. Transfer learning could be understood as a methodology in which a previously learned model is reused on a new problem. The program implemented uses the knowledge learned from a prior task or training and increases the prediction accuracy about the new task.

Here, we utilized resnet34 CNN pre-trained models, these are pre-trained models on ImageNet datasets, best used case for classification purpose. In more layman terms, an already trained machine learning model is transferred to a different closely linked problem throughout transfer learning. Although it requires high CPU power, it is still the best technique to learn as the previous task acts as an updated feature for the new data, improving the decision-making to detect the presence of disease in a new unknown eye image. Since the dataset is imbalanced, we used Cohen's quadratically weighted Kappa as it provides better metrics to deal with such imbalanced datasets by assisting a better methodology for measuring inter-rater agreement for categorical classification. After batch processing, the loss as the learning rate increases, we found loss decreased in Figure 2 gradually.

After implementing the batch processing, we compared two rounds of learning optimizations and the difference in loss after batch processing was completed. Figures 3 represent the loss, which decreases further as the batch processing continues in two stages and an effective transfer learning process is carried out.

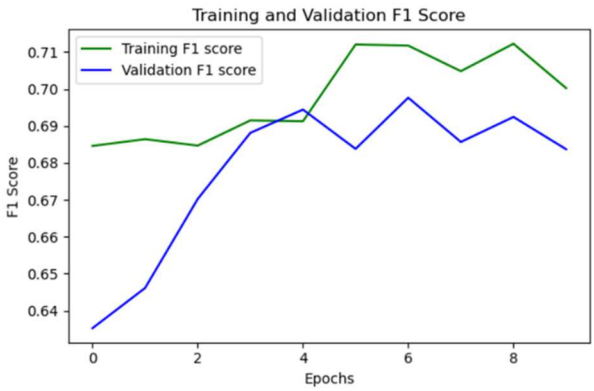

Fig 3. First Phase of Batch Processing

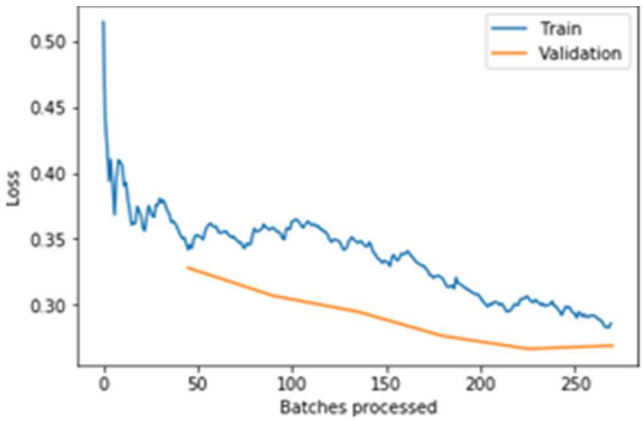

Fig 4. F1 value plot Validation and Training F1 Scores show improvement over the epochs

Figure 3 represent the loss after first stage of batch processing on processed data. We found that loss gradually kept decreasing, which was desired for the training and testing. It is because of the transfer learning methodology, which used every single training data as a feature of the next phase of training and, at last, which served as an abundance of features understanding in a more efficient way, which brings better accuracy for the detection of disease. Further we checked the

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accuracy of training and validation which was averaged around 73.83 % and 71.96 % respectively. Recall parameter a mathematical parameter mostly used for evaluating the performance of machine learning models was averaged around 74 and 71 for training and validation resp.

The mathematical equation for recall:

Recall = TP/TP+FN

Where, TP = True Positives

FN = False Negatives

Further, F1 values are an important parameter for image analysis models [10]. On analysis validation and training F1 scores are found to show improvements suggesting the models can be used with further refinements.

In addition to our effort to develop this model, we utilized Test Time Augmentation (TTA) to improve the model's score. TTA is a methodology implemented when a fit model is used to predict test or unseen data. TTA involves creating multiple slightly modified copies of examples in the dataset. TTA model improves the accuracy of the model. Previous studies have also predicted early-stage lung cancer and breast cancer using machine learning and deep learning (11,12)

IV. Conclusions

The medical condition diabetic retinopathy diagnosis is studied by using artificial intelligence (AI) and deep learning methods. The convolutional neural networks (CNNs) model used retinal pictures for classification and identification of DR. The use of AI and deep learning usage by medical professionals for diabetic retinopathy diagnosis may prove to be quite advantageous. Patient care can be improve as a informed desicion can be taken by medical practitioners about therapy and, early diagnosis. The AI and deep learning combination for the detection of DR could become a game changer for working professionals in medical sciences. The strategy may lower the work pressure on healthcare systems and improve the effectiveness for controlling DR. Further, to make the model robust and perfect investigations should be done on model, examining its suitability for other retinal ailments, and integrating the computerized system into medical procedures. Alarming progress in artificial intelligence and deep learning holds great promise for improving DR diagnosis and treatment, leading to better patient outcomes and more efficient healthcare delivery.

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